Independent PyLith Reproduction of Axial Seamount Failure

Axial Modeling Project

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Abstract

Cabaniss et al. (2020) used three-dimensional finite elements to relate Axial Seamount's bottom-pressure-recorder (BPR) deformation history to magma reservoir pressure, host-rock stress, and eruption timing. This project is rebuilding that workflow independently in PyLith, replacing the paper's COMSOL implementation while using its written equations, parameters, and reported outcomes as comparison targets. The central reproduction target is the 12–14 MPa eruption overpressure reported for temperature-dependent host models, together with the associated tensile failure and connected Mohr–Coulomb path. The current implementation runs elastic, Maxwell, thermal, and failure postprocessing cases and calibrates event-window pressure from Center BPR observations. Numerical compatibility checks cover a spherical Mogi source, thermal transfer, mesh sensitivity, and basal spring mechanics. The target failure threshold has not been reproduced. Ellipsoid compliance remains mesh-sensitive; production pressure fits use a fixed base; the finite Winkler solve has a large accepted residual; and failure-path onset changes with time resolution. Maxwell branch properties, the temperature-dependent modulus law, friction notation, and tensile strength are unresolved. These results define the work required to turn the bounded PyLith implementation into an independent reproduction.

1 Reproduction target

The target study models 22 years of Axial deformation with a pressurized ellipsoidal reservoir and four host rheologies. It calibrates pressure from the Center BPR history and evaluates reservoir tensile failure together with a through-going Mohr–Coulomb path. The temperature-dependent cases report eruption at 12–14 MPa overpressure. The paper and its supplement specify the model and provide numerical comparison values; author code, COMSOL files, and author-generated fields are not used.

The project uses a 6 by 3 by 1 km ellipsoid centered 1.6 km below the seafloor and oriented N30°W. The 50 by 50 by 10 km domain is a documented project assumption because the paper does not state its model-box dimensions. The model applies cavity pressure as a boundary traction, uses lateral rollers, and calculates a steady thermal field for the temperature-dependent material cases. The paper's base condition is a Winkler foundation; production pressure calibrations still use a fixed base while the finite-spring implementation is being stabilized.

2 Implemented calculations

PyLith reads the Gmsh ellipsoid mesh and writes HDF5 displacement and Cauchy stress fields. The project interpolates the saved surface displacement at the Center BPR location, estimates a pressure history for each rheology, and postprocesses saved stress states for tensile failure and face-connected

Mohr–Coulomb paths. Current calibrations cover separate 1998 and 2011 event windows. Both include observed eruption deflation, so their failure histories are retrospective diagnostics rather than independent predictions of event timing.

The present temperature-dependent modulus map decreases linearly from 50 GPa at 0 °C to 20 GPa at 1200 °C. This is an explicit project assumption: the printed Eq. 16 instead increases modulus with temperature, contrary to the text's brittle and ductile labels. Generalized Maxwell branch fractions and relaxation times are also incomplete in the written source, so current branch values are synthetic. The failure calculation uses a 1 MPa cohesion, 25° friction angle, and zero pore pressure as a diagnostic proxy; the source's friction notation and tensile strength remain unresolved.

3 Numerical checks and current results

The bounded Mogi benchmark has not converged under mesh refinement. Its 2,995-element case reports a 44.9% vector-field error and a 55.9% center error relative to the analytical solution. A nearby refined case reports 42.7% and 54.2%, respectively. The meshes are nonnested, and the error trend does not establish convergence. The ellipsoid unit response also varies substantially among bounded meshes, preventing a stable pressure scale.

The finite-spring diagnostic uses a regional stiffness prior of 32,373 Pa/m and a depth-averaged density approximation. Its nonlinear solver accepts an initial residual of 5.40×10^3 under a 1.00×10^4 tolerance at iteration zero. A tighter solve previously failed in KSP. Production calibrations therefore retain a fixed base, which does not match the written boundary condition. Thermal mesh checks show sampled temperature differences up to 324 °C near the reservoir on a nonnested mesh sequence.

A static failure screen with the current strength proxy finds no connected reservoir-to-surface path at the 12–14 MPa comparison range for either basal treatment. Time-step refinement also changes saved path onset and repeated path appearance, so the current threshold is neither stable nor a reproduced eruption prediction.

4 Work required for reproduction

First, resolve the printed modulus conflict and the friction notation against the publisher-linked supplement and cited methods. Table S1 gives a 25° friction angle; the dimensionless Eq. 25 coefficient is tan(25°). Second, establish mesh and domain convergence for Center displacement, cavity-adjacent stress, and the analytical benchmarks. Third, stabilize the finite Winkler boundary and use it in the pressure-calibrated model. Fourth, recover a defensible Maxwell spectrum and a continuous Center BPR forcing history without fitting the reported failure pressure. Finally, refine time integration until the joint tensile and connected-shear criterion yields a stable threshold that can be compared with 12–14 MPa and the reported event times.

5 Reproducible record

The complete two-cycle sequence is not yet implemented. The command `make reproduction-status` lists source and model requirements, while `make reproduce` stops without generating a claimed paper result. Eight available processed Center archive files have been exported without joining deployment datums or assuming the paper's exact channel. The current coarse cavity mesh has a 34.90% geometric volume deficit against the ideal 6 by 3 by 1 km ellipsoid. Its volume-change reader is a software check,

not a reproduced history. Raw observations, processed BPR series, meshes, and PyLith output remain local. The source audit, parameter record, and figure status are in the repository's `docs/` directory.

References

[1] H. E. Cabaniss, P. M. Gregg, S. L. Nooner, and W. W. Chadwick, "Triggering of eruptions at Axial Seamount, Juan de Fuca Ridge," *Scientific Reports*, vol. 10, 10219, 2020. doi:10.1038/s41598-020-67043-0.